

Parameterizing Pathological Gait with Limit Cycle Oscillators: From Pure Phase to Full Amplitude Resilience

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Abstract

Continuous biological motions like human walking represent stable limit cycles, yet distinguishing the subtle structural instabilities of pathological gait (e.g., following a stroke) from natural variations due to walking speed remains a challenge. In this note, we compile two specific phenomenological modeling approaches applied to empirical high-dimensional marker trajectories: an initial phase-only oscillator model (based on the Kuramoto/Shaevitz framework) and a full amplitude-phase Stuart-Landau model. By characterizing both the global cycle stability and the localized step-to-step volatility of spatial parameters, we uncover differences between Able-Bodied (AB) individuals and Hemiparetic subjects.

1 Phase-Only Oscillators: The Kuramoto Framework

Initially, we approached the cyclic motion via a purely phase-based framework. Human marker coordinates were embedded into a delay-coordinate Hankel matrix and projected via Singular Value Decomposition (SVD) to identify the two primary dominant harmonic cycles. The phase angles of these modes were fit to a coupled oscillator framework to evaluate the temporal coupling between joints.

Figure ?? demonstrates the model tracking phase progression. By exploring the parameters driving this timing constraint, we constructed a UMAP projection of the individual human trials (Figure ??). While the model accurately parameterizes the timing characteristics of the limit cycle, subsequent analysis of the “phase resilience”—a global timing-recovery metric—revealed strong confounding correlations with the subject’s self-selected walking speed (Figures ?? and ??). This suggested that while stroke pathways disrupt gait, modeling pure timing excludes critical information about spatial instability and Floquet multiplier structure.

2 Encoding Amplitude: The Stuart-Landau Model

To capture structural spatial limit-cycle phenomena, we upgraded to a Stuart-Landau (Hopf) network, tracking the complex amplitudes $z(t)$:

$$\frac{dz_i}{dt} = \mu_i z_i - z_i |z_i|^2 + i\omega_i z_i + \sum_j K_{ij} z_j$$

A central benefit of this model is the direct accessibility of the structural dampening parameter μ , reflecting Floquet-like decay toward the attractor. Fitting dynamic experimental gradients without unbounded derivative blowups required enforcing strong biologically constrained empirical frequencies ($\omega \approx \omega_{actual}$) and strict Ridge regularization on coupling magnitudes K_{ij} evaluated via central differences. The globally fitted parameters correctly integrate autonomously to replicate the fundamental harmonics of observed human gait modes (Figure ??).

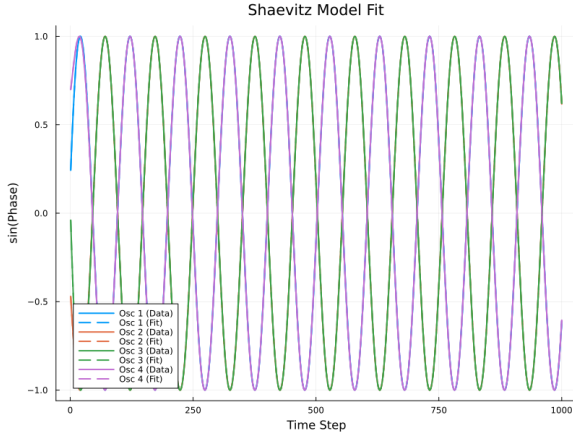


Figure 1: Coupled phase oscillator fit.

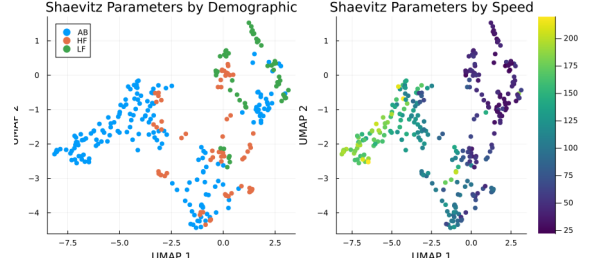


Figure 2: UMAP clustering of the phase-only parameter regime across clinical populations.

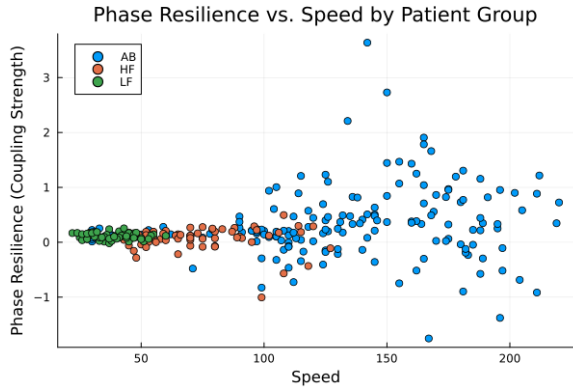


Figure 3: Phase resilience tightly tracks simple walking speed rather than pathology alone.

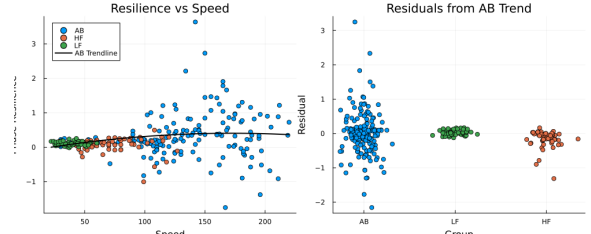


Figure 4: Residuals of phase resilience upon factoring out speed.

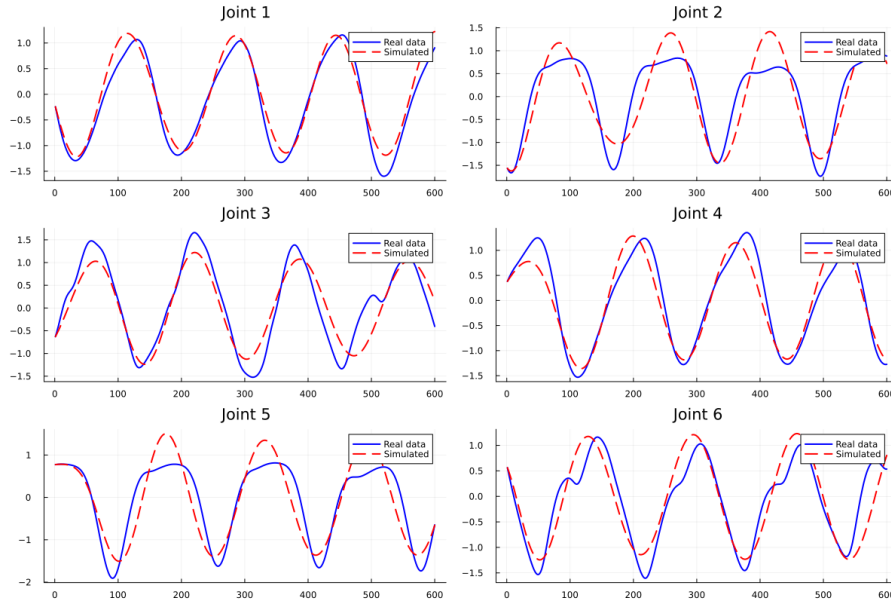


Figure 5: Autonomous forward simulation of the regularized Stuart-Landau fit successfully tracking biological multi-channel amplitude variations over time.

2.1 Global Resilience Correlates

Computing the global structural resilience metric (μ) confirmed strong dependence on velocity. Utilizing a non-parametric localized LOESS regression exclusively on Able-Bodied users, we mapped the baseline “healthy” limit-cycle rigidity across kinematic walking speeds.

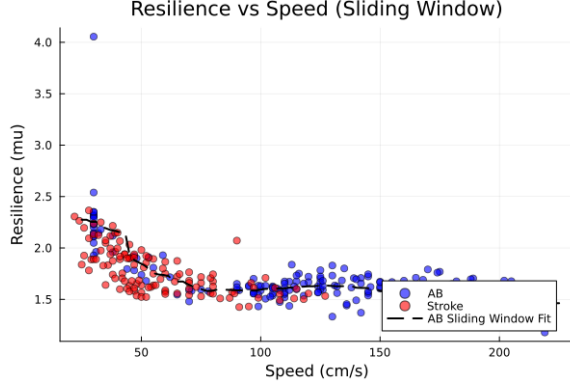


Figure 6: Spatial structural rigidity μ vs. Velocity. Note the sliding-window baseline characterizing healthy subjects (AB).

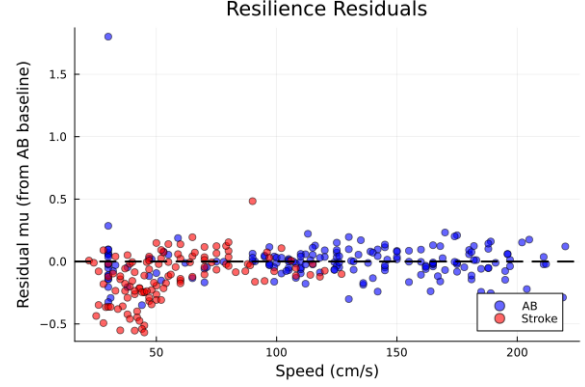


Figure 7: Residual structural resilience of Stroke groups relative to the healthy non-parametric baseline.

3 Intra-Trial Volatility: The Step-by-Step Variance

While the global parameters highlighted a systematic deficit in stroke patients (reflecting a clinical resilience delay of roughly $\sim 20\text{ms}$), such a minor average margin might not capture the true mechanism of pathological locomotor instability. We hypothesized the deficit operates dynamically: stroke patients frequently shift attractor stabilization strategies step-by-step to compensate for motor noise, contrasting with the uniform, fixed-rigidity cycles of healthy pedestrians.

By executing the Stuart-Landau inversion across binned temporal windows (e.g., estimating local μ parameters using 300 temporally consecutive tracking frames per step), we quantified intra-trial deviations. Rather than evaluating standard residuals, we quantified the inherent variation of μ over time. We plotted the intra-trial statistical volatility (Standard Deviation) of the stability corrections over time against local velocities (Figure ??) and distributed by condition (Figure ??).

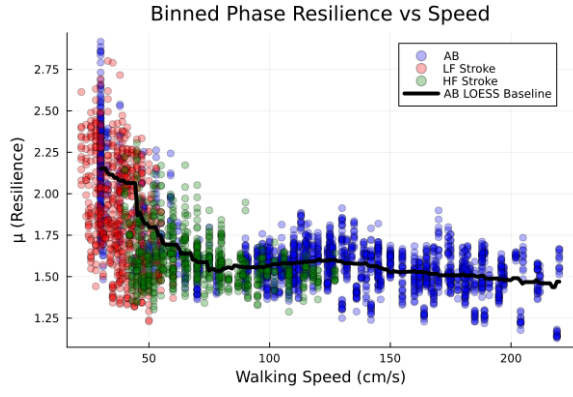


Figure 8: Binned spatial rigidity evaluation over single-step local temporal sliding windows.

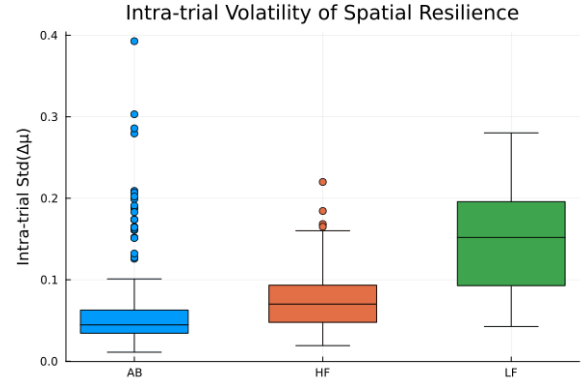


Figure 9: Intra-trial Volatility of Spatial Resilience across populations. Pathological patients display significantly rougher dynamic recalibration from step to step.